

unbound: forward-addr directives placed outside forward-zone block due to heredoc formatting

Component: Services: Unbound DNS

Version: OPNsense 26.1.5 (confirmed present in master as of 2026-03-27)

Severity: High — Unbound ignores all configured upstream forwarders

File: src/etc/inc/plugins.inc.d/unbound.inc

Lines: 307-323

Summary

In `unbound_generate_config()`, the PHP heredoc generating the forward-zone block for the root zone (.) contains a blank line between `'name: "."` and the EOD; terminator. Because forward-addr lines are appended to `$forward_conf` after the heredoc closes, the blank line causes them to appear outside the forward-zone block in the generated `unbound.conf`. Unbound silently ignores forward-addr directives that appear outside a zone block, resulting in no upstream forwarders being configured despite GUI settings.

Environment

OPNsense 26.1.5 (FreeBSD 14) | Unbound 1.24.2 | Query Forwarding enabled | System nameservers: 1.1.1.1, 8.8.8.8

Steps to Reproduce

1. Configure Unbound with Query Forwarding enabled and upstream nameservers
2. Run: `configctl unbound restart`
3. Inspect: `grep -A5 'forward-zone' /var/unbound/unbound.conf`
4. Observe blank line between `name: "."` and forward-addr directives

Expected Result

```
forward-zone:
  name: "."
  forward-addr: 1.1.1.1
  forward-addr: 8.8.8.8
```

Actual Result

```
forward-zone:
  name: "."

  forward-addr: 1.1.1.1
  forward-addr: 8.8.8.8
```

Blank line breaks the zone block. forward-addr parsed as outside the zone.

Root Cause

Lines 307-323 in `unbound_generate_config()`:

```

$forward_conf .= <<<EOD
# Forwarding
forward-zone:
    name: "."

                                <-- blank line here

EOD;
foreach ($dnsservers as $dnserver) {
    $forward_conf .= "\tforward-addr: $dnserver\n";
}

```

The blank line is written literally into unbound.conf, breaking zone block continuity. Additionally, \

Impact

Any deployment using Query Forwarding has a silently broken forward-zone configuration. Unbound starts without error but ignores all configured upstream forwarders. DNS falls back to recursive resolution from root servers, which may also fail depending on network topology. Combined with Bug #2 (chroot), this results in complete DNS failure.

Fix

Replace heredoc with string concatenation:

Before:

```

$forward_conf .= <<<EOD
# Forwarding
forward-zone:
    name: "."

```

EOD;

```
$forward_conf .= "\tforward-addr: $dnserver\n";
```

After:

```

$forward_conf .= "# Forwarding\nforward-zone:\n    name: \".\"\n";
$forward_conf .= "    forward-addr: $dnserver\n";

```

Validation

```

$ grep -A5 'forward-zone' /var/unbound/unbound.conf
forward-zone:
    name: "."
    forward-addr: 1.1.1.1
    forward-addr: 8.8.8.8

```

```

$ unbound-control list_forwards
. IN forward 1.1.1.1 8.8.8.8

```

Suggested Commit Message

unbound: fix forward-zone heredoc formatting causing forward-addr outside zone block

Note on Bug Interaction

All three bugs (this report, #1 Python module, #2 chroot) are present simultaneously in OPNsense 26.1.5 and interact to produce complete DNS failure. Any one bug alone may not cause total failure in all environments, but their combination produces a total DNS outage that is extremely difficult to diagnose because each failure mode masks the others. Root cause analysis required WAN packet capture, Unbound infrastructure cache inspection, and PHP source code review to isolate.

Discovered by Alec Sanchez during OSIM framework homelab commissioning, 2026-03-27. Contact: secureadminpublic@protonmail.com | github.com/osim-framework